HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Qualified **SSD and HDD drive solutions** on Dell server products, collaborating with internal development partners to prepare drive programs for factory and customer readiness.
- Addressed complex design challenges on emerging technologies through hands-on technical experience, focusing on **NVMe and PCIe** interfaces.
- Gained expertise in **drive technology solutions**, encompassing hardware, firmware, and software interactions across enterprise storage systems.
- Developed **Python scripts** to automate data extraction and analysis for drive performance testing, reducing manual effort by approximately **40%**.
- Performed **digital hardware debug** on complex systems, working with and influencing global engineering teams to resolve critical issues.
- Managed **x86 Server Architecture design** considerations, ensuring compatibility and optimal performance for integrated drive solutions.
- Containerized **drive testing applications** using **Docker**, enabling consistent deployment across various testing environments on **AWS EC2** instances.

SKILLS

Hardware & Firmware – HDD, SSD, NVMe, PCIe, Drive Technology

Server Architecture – x86 Server Architecture

Software Development – Java, Spring Boot, Python, REST APIs, Docker

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Jenkins, Git

Databases & OS – SQL (PostgreSQL, MySQL, Oracle), Linux

Testing & Debugging – Digital Hardware Debug, System Design, Validation

PROJECTS

Enterprise Drive Performance Validation Suite

- Engineered a Python-based framework to automate the execution of drive performance tests on **Dell server hardware**, integrating with **AWS Lambda** for distributed execution.
- Developed custom scripts to parse and analyze test logs, identifying performance bottlenecks and generating summary reports that highlighted key regressions for engineering review.
- Implemented **REST APIs** using Flask to allow remote triggering of test suites and retrieval of results, streamlining the validation process for remote teams.

SSD Firmware Update and Diagnostics Tool

- Built a command-line utility in **Java** to manage firmware updates for **SSD drives** across a fleet of test servers, incorporating rollback capabilities.
- Integrated diagnostic checks using **SQL queries** against a PostgreSQL database to verify drive health and firmware compatibility before and after updates.
- Designed the application with modular components for future expansion to support **HDD drives** and additional diagnostic routines.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)